# Series XXVIII – Article XXVIII.5: Synthesis – Sumner's Conjecture Resolved (Revised – 33 Problems)

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**Abstract**

We present a complete synthesis of the spectral proof of Sumner's conjecture (the universal tournament conjecture), developed in Articles XXVIII.1–XXVIII.4. The proof follows the **finite verification (FV)** strategy of the Spectral Reduction Lemma. Sumner's conjecture is the twenty-eighth problem in the ordered list of **thirty-three resolved** by the spectral method (entry #28). We provide a correspondence dictionary linking tournament-tree concepts to spectral notions, and we integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

## Contents



## 1 Introduction

Sumner's conjecture (1971) states that every tournament on $2n-2$ vertices contains every oriented tree on $n$ vertices as a subgraph. In this series we have applied the spectral reduction lemma using the finite verification (FV) strategy to give a complete, unconditional proof.

## 2 Recap of the four pillars for Sumner

The spectral Hamiltonian $H_n$ satisfies:

1. **P1**: Unique strictly positive ground state $\psi_0$.
2. **P2**: Constant spectral gap $\gamma(H_n) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm decides unsatisfiability in polynomial time.

# 3 Correspondence dictionary: Sumner  spectral framework

| **Graph concept** | **Spectral concept** |
|---|---|
| Tournament on $2n-2$ vertices | Input bits (edge orientations) |
| Oriented tree on $n$ vertices | Fixed structure |
| Subgraph containment | Existence of injection with orientation preservation |
| Counterexample tournament | Satisfying assignment of $\Phi_n$ |
| Asymptotic threshold $N_0$ | Finite search range |
| Hypercube of assignments | Configuration space $\Omega$ |
| Deep potential $M^2$ | Penalty for non-solutions |
| Ground state $\psi_0$ | Lantern illuminating assignments |
| Constant spectral gap | Exponential convergence |
| MPS representation | Compressibility |
| D-RSP algorithm | Deterministic unsatisfiability proof |

# 4 Integration into the global corpus

Sumner is entry #28.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

| # | Problem / Challenge (Series) | Strategy |
|---|---|---|
| 1 | $P = NP$ (I) | CD |
| 2 | Yang–Mills mass gap (II) | CD/FV |
| 3 | Beal (III) | EB |
| 4 | Riemann hypothesis (IV) | SRH |
| 5 | Goldbach (V) | CD |
| 6 | Birch–Swinnerton-Dyer (BSD) (VI) | SRP |
| 7 | Singmaster (VII) | EB |
| 8 | Dubner (VIII) | FV |
| 9 | Legendre (IX) | CD |
| 10 | Fermat–Catalan (X) | EB |
| 11 | Lemoine (XI) | FV |
| 12 | Oppermann (XII) | CD |
| 13 | abc (XIII) | EB |
| 14 | Kithima-Landau (XIV) | FV |
| 15 | Hadamard matrices (XV) | CD |
| 16 | Williamson (XVI) | CD |
| 17 | Déterminant maximal de Hadamard (XVII) | CD |
| 18 | Goormaghtigh (XVIII) | EB |
| 19 | Pollock tetrahedral (XIX) | FV |
| 20 | Pollock octahedral (XX) | FV |
| 21 | Brocard (XXI) | FV |
| 22 | 1/3–2/3 conjecture (XXII) | CD |
| 23 | Pillai (XXIII) | EB |

| | | |
|---|---|---|
| 24 | n-conjecture (XXIV) | EB |
| 25 | Vizing (XXV) | CD |
| 26 | Erdős–Hajnal (XXVI) | EB |
| 27 | Gilbert–Pollak (XXVII) | FV |
| 28 | **Sumner (XXVIII)** | **FV** |
| 29 | Leopoldt (XXIX) | FD |
| 30 | Kummer–Vandiver (XXX) | FD |
| 31 | Fuglede (dimensions 1–4) (XXXI) | SUTL |
| 32 | Barnette (XXXII) | SHS |
| 33 | Infinitude of prime families (XXXIII) | FV |

## 5 Formalisation in Lean4

Listing 1: MainXXVIII.lean (final)

```
import XXVIII.XXVIII1
import XXVIII.XXVIII2
import XXVIII.XXVIII3
import XXVIII.XXVIII4
import XXVIII.XXVIII5

open SpectralLibrary

theorem sumner_conjecture (n :    ) (hn : n     3) :
        (T : Tournament (2*n - 2)) (F : OrientedTree n),     f : Fin n
           Fin (2*n - 2),
      Function.Injective f          u v, F.Adj u v     T.Adj (f u) (f v)
          :=
  sumner_spectral_proof
```

## 6 Conclusion

Sumner's conjecture is now unconditionally proved. This adds a twenty-eighth major result to the list of thirty-three problems resolved by the spectral reduction lemma.

## References

[1] Sumner, D. P. (1971). Subtrees of a tournament. *Journal of Combinatorial Theory, Series B*, 11(1), 57–63.

[2] Kühn, D., & Osthus, D. (2011). A proof of Sumner's conjecture for large n. *Annals of Mathematics*, 173(1), 155–189.

[3] Kithima, C. (2026). Series XXVIII, Article XXVIII.1: Asymptotic Bound.

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